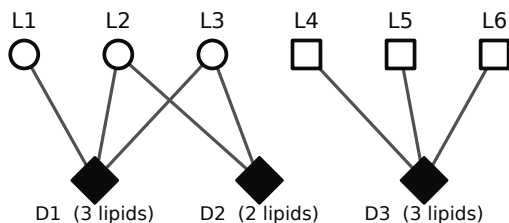
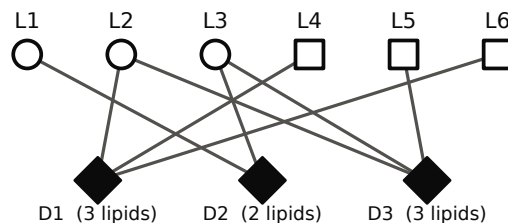
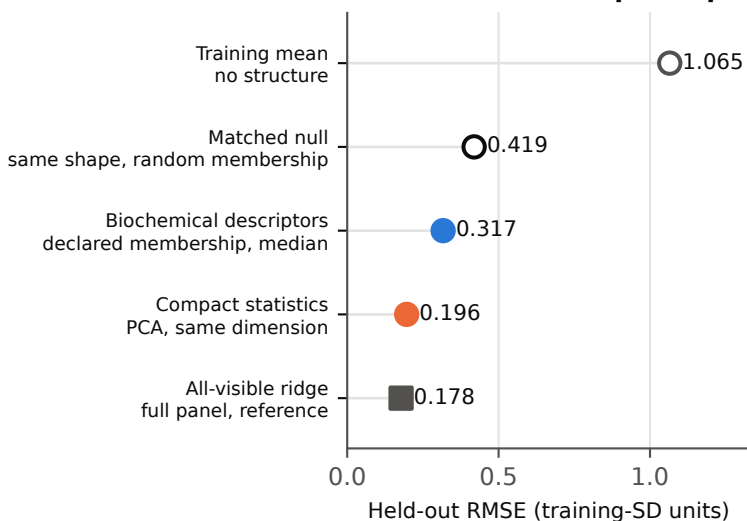
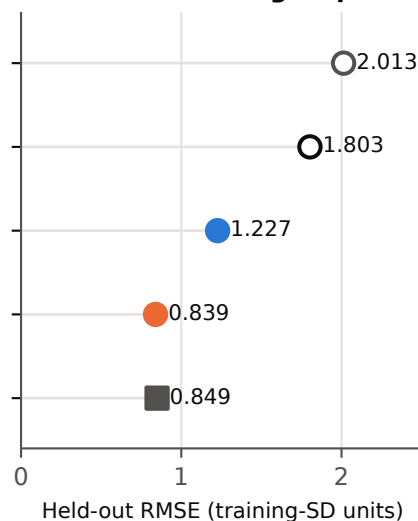


A**Declared biochemical membership****Matched null: same shape, rewired membership**

The null keeps every descriptor's size and every lipid's number of descriptors; only which lipids share a descriptor is randomized. Circles and squares mark two lipid classes: declared descriptors group like with like, the null mixes them.

B ST002081 · 112 participants**C ST000818 · 15 groups**

Lower is better. Matched null and biochemical descriptors share shape, so their gap isolates biochemical membership.

D Both steps hold in every dataset